SIS SAMPLE DRAFT — FULL TIER

Richland County Sample — Shopping Center / Retail

Traffic Impact Study

Prepared August 8, 2026 · 34.0605, -80.9670

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EXECUTIVE SUMMARY

This Traffic Impact Study evaluates the proposed Richland County Sample — Shopping Center / Retail within Columbia (SC) MSA, South Carolina (Columbia region (Richland / Lexington)), in accordance with SCDOT's Access and Roadside Management Standards (ARMS, 2008 edition, updated July 8, 2025) Chapter 6 and Traffic Engineering Guideline TG-21. In South Carolina the TIS is an element of the SCDOT encroachment-permit process: the study scope must be established with the District Traffic Engineer (DTE) before analysis begins, and the DTE makes the final mitigation determination.

2	680	1	0
INTERSECTIONS	PM PEAK TRIPS	AT LOS E/F	LOS DROPS

1.0 INTRODUCTION

Project name	Richland County Sample — Shopping Center / Retail
Land use	LU 820 — Shopping Center / Retail
Development size	85 1,000 sqft GLA
Site coordinates	34.0605°, -80.9670°
Build-out year	2027
Controlling jurisdiction	Columbia region (Richland / Lexington)
MPO / study	COATS (Columbia Area Transportation Study), staffed by CMCOG
Programmed projects source	SCDOT FFY 2024–2033 STIP, as amended (e-STIP)
Submittal channel	SCDOT EPPS (Encroachment Permit Processing System) via the District Permit Engineer

1.1 TIS Applicability

Accepted Columbia-area submittals follow "City of Columbia and SCDOT guidelines" — i.e., ARMS Chapter 6 with the study area established with City planning staff. The statewide 100 peak-hour-trip ARMS trigger governs on SCDOT facilities.

The proposed development's estimated peak-hour generation of 680 trips meets or exceeds the ARMS 100 peak-hour-trip threshold; a TIS is required for the encroachment permit.

2.0 PROJECT TRAFFIC

ARMS Chapter 6 calls for trip generation per the latest edition of the ITE Trip Generation Manual, with internal-capture, pass-by, and transit reductions justified to the DTE. This screening analysis applies public-data trip-generation rates (NHTS 2017 / SANDAG 2002 / NCHRP 716), with every rate tagged so the jurisdiction-approved ITE figure can be substituted at submittal.

Daily trips (gross)	6,800
AM peak hour	166 in / 106 out
PM peak hour	326 in / 354 out
Counts standard	AM and PM weekday peaks minimum; counts ≤12 months old, school in session, seasonally adjusted

3.0 TRAFFIC VOLUME DEVELOPMENT AND CAPACITY ANALYSIS

Scenarios per ARMS: existing year, build-out no-build, and build-out build. Capacity analysis follows the Highway Capacity Manual (HCM 6th Edition consistent) with LOS reported for all approaches and movements; coordinated signal systems are analyzed as a system.

3.1 TG-21 Level of Service Standard — LOS C

Per SCDOT Traffic Engineering Guideline TG-21 (Mitigation of Traffic Impacts), the acceptable level of service for the design (peak) hour is LOS C or better for all roadway types statewide, applied in lieu of locally preferred thresholds. Where the baseline already operates at or below LOS C, the baseline LOS must be maintained or improved; where the baseline is LOS F in a congested urban area, mitigation is at the DTE's determination. Note: this is a stricter standard than the LOS D convention applied in several neighboring states.

No.	Intersection	Exist.	No-Build	Build LOS/Delay	TG-21 (LOS C) check
1	Two Notch Road & Shakespeare Road	B	B	B / 16.2s	meets LOS C
2	Near Two Notch Road	F	F	F / 120.0s	baseline maintained

4.0 FINDINGS, ACCESS MANAGEMENT, AND SIGNAL WARRANTS

0 intersections project a LOS drop and 1 operate at LOS E or F under Build conditions; mitigation alternatives (turn lanes with storage per ARMS Ch. 5 and the SCDOT Roadway Design Manual Ch. 9, signalization, or operational improvements) are developed with the DTE, who makes the final determination.

Access management: ARMS requires demonstration that the fewest driveways necessary serve the site, with sight-distance verification at each access point. Signal proposals require a signal warrant analysis per the MUTCD. Both are site-plan-dependent and are prepared at submittal.

5.0 ARMS TECHNICAL COMPLETENESS CHECKLIST

SCDOT returns incomplete studies unreviewed. The ARMS completeness checklist is tracked below against this screening draft; unmet items are produced at formal submittal.

Checklist item	Status in this draft
SC PE stamp and signature	At submittal (screening draft is unsealed)
Introduction + executive summary	Included
Existing conditions (classification, peaks, LOS, volumes)	Included (screening level)
Trip generation / distribution / LOS analysis	Included (public-data rates, tagged)
Crash data / speeds / sight distance	At submittal (site-specific)
Mitigation w/ turn-lane storage lengths	Indicated per TG-21 table; design at submittal
Figures (vicinity, site plan, volumes, lane configs, LOS)	Volume/LOS figures included; site-plan figures at submittal
Tables (trip gen, LOS existing/background/build/mitigated)	Included (screening level)
Technical appendix (capacity reports, warrants)	Capacity worksheets appended; Synchro/ warrants at submittal

This TIS is to be prepared under the direct charge of and sealed by a registered South Carolina Professional Engineer with expertise in traffic engineering (ARMS), with seal, signature across the seal face, and date per S.C. Code §40-22-270 and Regs. Ch. 49, and the firm's Certificate of Authorization. Electronic sealing is accepted in current practice.

SCREENING DISCLOSURE: this document is a screening-level draft prepared before DTE scoping. Scope, growth rates, background developments, and mitigation are established with the District Traffic Engineer; a submittal-grade analysis supersedes this draft.

FOUR-STEP TRAVEL DEMAND MODEL

Off-site project trips are distributed and assigned with the four-step urban transportation modeling process (FHWA; NCHRP Report 716, Travel Demand Forecasting). Each step below shows what the engine computed for this site.

Step 1 — Trip Generation

Public-data screening trip rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) applied to the proposed land use give the site's productions.

Land use	LU 820 — Shopping Center / Retail
Size	85 1,000 sqft GLA
Daily trips (productions)	6,800
PM peak-hour trips	680 (326 in / 354 out)

Step 2 — Trip Distribution (Gravity Model)

A production-constrained gravity model allocates trips to surrounding signals: $T_j = P \cdot (A_j \cdot F_j) / \Sigma (A_x \cdot F_x)$. Attractiveness A_j is each signal's through-volume; the friction factor F_j is the NCHRP-716 gamma function $F = a \cdot t^b \cdot e^{(c \cdot t)}$ (home-based-work: a=28507, b=-0.02, c=-0.123) on the travel time t to the signal.

Signal (attraction zone)	Dist mi	Time min	PM trips	Share
Two Notch Road & Shakespeare Road	0.33	0.8	87	66.9%
Near Two Notch Road	0.47	1.1	43	33.1%

Top 12 distribution zones by assigned PM-peak trips shown; the full set is in the capacity appendix.

Step 3 — Mode Choice

Auto-mode share 94% applied via a binary logit $P(auto) = 1 / (1 + e^{-(ASC - \lambda \cdot \Delta GC)})$ calibrated to the metro's measured auto-mode share (ACS B08301) and shifted by site urbanity (local density), so a denser, more transit-served site splits further from auto than a greenfield parcel in the same metro. The remaining 6% of trips (transit, walking, cycling) do not load the off-site roadway network; only auto trips are carried into Step 4.

Step 4 — Route Assignment

A capacity-constrained assignment using the BPR volume-delay function $t = t_0 \cdot [1 + 0.15 \cdot (v/c)^4]$ iteratively shifts trips away from over-capacity signals toward less-congested alternatives until the assignment is congestion-consistent. Highest post-build v/c among study signals: 1.27. Per-signal v/c, delay, LOS, and queue are in the capacity appendix. (Road-network corridor routing was not available for this region; per-signal capacity-constrained assignment was used.)

APPENDIX — INTERSECTION CAPACITY ANALYSIS WORKSHEETS

One worksheet per study intersection: a turning-movement diagram for each analyzed peak period plus the per-approach HCM capacity table. Analysis follows the Highway Capacity Manual 6th Edition signalized-intersection method; v/c, control delay (sec/veh), Level of Service, and 95th-percentile back-of-queue (ft) are reported for the Existing (No-Build) and Build conditions.

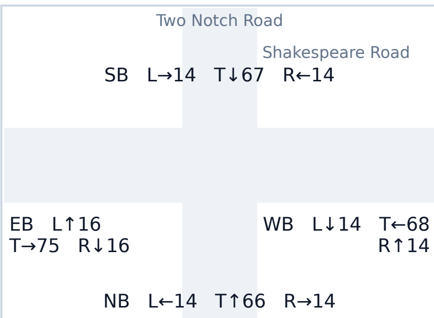
Background turning-movement volumes in the diagrams are distributed from each approach total using an estimated 15/70/15 (Left/Through/Right) split. Project-trip movements are assigned geometrically from the study's directional trip distribution (see each worksheet's Affected movements table). Replace both with measured turning-movement counts (TMCs) before a formal submittal.

A.1 Two Notch Road & Shakespeare Road

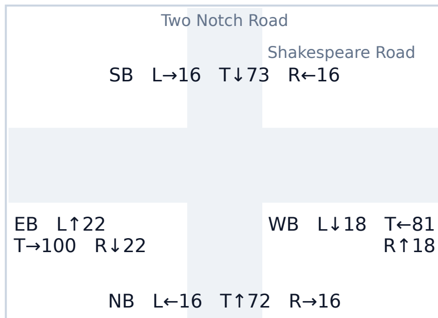
Signal ID	columbia-sc-511
Location	Central Columbia (SC) · 0.33 mi from site
Added PM peak trips	87
Existing / No-Build (PM)	LOS B · 15.7 s/veh · v/c 0.21
Build (PM)	LOS B · 16.2 s/veh · v/c 0.26
Δ control delay	+0.5 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

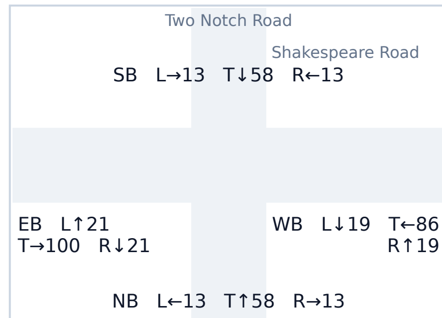
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	104	0	104	0.13 → 0.13	14.8 → 14.8	B → B	63
SB	105	0	105	0.13 → 0.13	14.8 → 14.8	B → B	63
EB	99	45	144	0.12 → 0.18	14.7 → 15.3	B → B	89
WB	75	42	117	0.09 → 0.14	14.4 → 14.9	B → B	71

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	45	52%
WB Through	42	48%

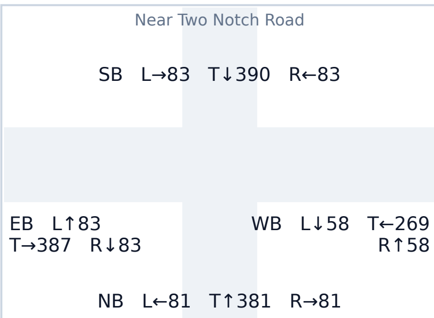
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.2 Near Two Notch Road

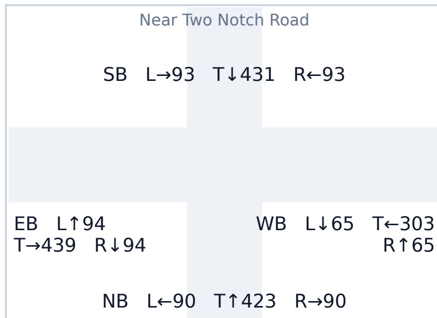
Signal ID	columbia-sc-580
Location	Central Columbia (SC) · 0.47 mi from site
Added PM peak trips	43
Existing / No-Build (PM)	LOS F · 120.0 s/veh · v/c 1.24
Build (PM)	LOS F · 120.0 s/veh · v/c 1.27
Δ control delay	+0.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)

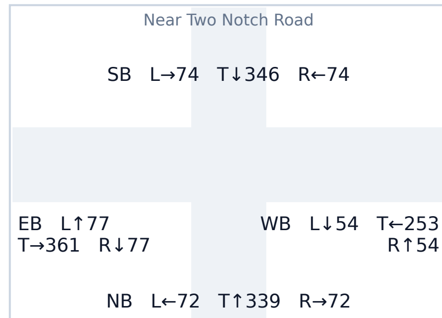
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	603	0	603	0.74 → 0.74	26.6 → 26.6	C → C	515
SB	617	0	617	0.76 → 0.76	27.4 → 27.4	C → C	533
EB	605	22	627	0.75 → 0.77	26.7 → 28.0	C → C	546
WB	413	21	433	0.51 → 0.53	19.9 → 20.4	B → C	324

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	22	51%
WB Through	21	49%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.